EXP.2. Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

Aim:

To model a 50 Ω microwave two-port network of length 50 mm, simulate its S-parameters (\(S_{11}, S_{12}, S_{21}, S_{22}\)) over the frequency range of 8–12 GHz, and calculate the return loss, insertion loss, and transmission efficiency at 10 GHz.

Software required:

Matlab

Program:

```
clc;
clear;
close all;

%% Input Parameters
Z0 = 50;              % Reference impedance (ohms)
Zc = 52;              % Characteristic impedance (ohms)
L = 50e-3;            % Length = 50 mm
f0 = 10e9;            % Operating frequency = 10 GHz
c = 3e8;              % Speed of light (m/s)
er = 2.2;             % Relative permittivity
loss_dB_per_m = 10;   % Assumed attenuation (dB/m)

%% Transmission Line Calculation
beta = 2*pi*f0*sqrt(er)/c;
alpha = loss_dB_per_m/8.686;
gamma = alpha + 1j*beta;

A = cosh(gamma*L);
B = Zc*sinh(gamma*L);
C = sinh(gamma*L)/Zc;
D = cosh(gamma*L);

%% ABCD to S-Parameters
den = A + B/Z0 + C*Z0 + D;

S11 = (A + B/Z0 - C*Z0 - D)/den;
S21 = 2/den;
S12 = 2*(A*D - B*C)/den;
S22 = (-A + B/Z0 - C*Z0 + D)/den;

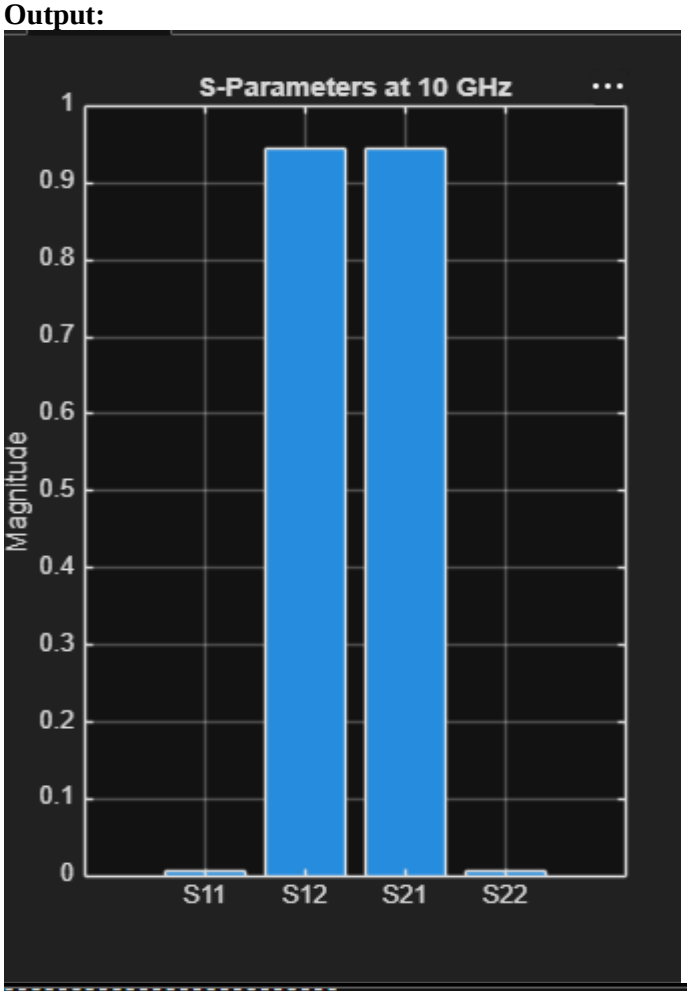
%% Performance Parameters
RL = -20*log10(abs(S11));
IL = -20*log10(abs(S21));
eta = abs(S21)^2*100;

%% Display Results
fprintf('PART A: RESULTS AT 10 GHz\n');
fprintf('------------------------\n');
fprintf('S11 = %.4f %+.4fj\n',real(S11),imag(S11));
fprintf('S12 = %.4f %+.4fj\n',real(S12),imag(S12));
fprintf('S21 = %.4f %+.4fj\n',real(S21),imag(S21));
fprintf('S22 = %.4f %+.4fj\n',real(S22),imag(S22));

fprintf('\nReturn Loss = %.4f dB\n',RL);
fprintf('Insertion Loss = %.4f dB\n',IL);
fprintf('Transmission Efficiency = %.4f %%\n',eta);

%% Plot S-Parameters
figure;
bar([abs(S11),abs(S12),abs(S21),abs(S22)]);
set(gca,'XTickLabel',{'S11','S12','S21','S22'});
ylabel('Magnitude');
title('S-Parameters at 10 GHz');
grid on;
```

Output:




```
S11 = 0.0032 -0.0060j
S12 = -0.9295 -0.1649j
S21 = -0.9295 -0.1649j
S22 = 0.0032 -0.0060j

Return Loss = 43.3379 dB
Insertion Loss = 0.5005 dB
Transmission Efficiency = 89.1140 %
>>
```